Supplementary Material — Forecasting Electricity Demand in Unseen Regions

This supplement accompanies the manuscript. All tables are generated by `src/generate_tables.py` from the metric files listed in `results/tables/table_manifest.csv`. Analyses marked *post-review* were added in response to an internal simulated review and are logged in DECISION_LOG.md (D14–D16).

1 S1. Full interval-calibration results

Table S1. Interval calibration on the test period, all settings (48 BAs).

Nominal	h (h)	Setting	Method	Mean coverage %	Min–max coverage %	Mean	cov. dev.	pp	BAs under-covered >5 pp	Median hourly width %
90%	1	in-region	QR	84.8	66.7–96.4	6.8	44	4.08	4.46	7.12
90%	1	in-region	SCP	88.3	54.8–99.5	7	31	4.8	4.96	7.45
90%	1	in-region	CQR	88.7	70.1–98.0	5.3	25	4.59	4.98	6.99
90%	1	in-region	SCP-T4	86.4	70.0–95.4	5.1	31	4.31	4.51	7.67
90%	1	in-region	ACI-CQR	90	89.9–90.1	0	0	4.76	5.34	7.15
90%	1	in-region	ACI-CQR-V	90	89.7–90.4	0.1	0	4.74	5.17	6.9
90%	1	unseen-region	QR	83.9	64.8–97.5	7.8	52	4.2	4.53	7.4
90%	1	unseen-region	SCP	87.5	52.6–98.9	7.2	31	4.78	4.93	7.72
90%	1	unseen-region	CQR	88.1	69.2–98.7	5.9	33	4.74	5.1	7.21
90%	1	unseen-region	SCP-T4	86.9	74.3–96.5	4.6	29	4.3	4.58	7.81
90%	1	unseen-region	ACI-CQR	90	89.9–90.2	0	0	4.9	5.49	7.4
90%	1	unseen-region	ACI-CQR-V	90	89.7–90.4	0.1	0	4.9	5.38	7.13
90%	24	in-region	QR	87.5	77.2–93.3	3.5	23	17.04	20.25	27.79
90%	24	in-region	SCP	89.1	71.7–99.3	5.5	31	20.29	20.96	31.43
90%	24	in-region	CQR	89.2	79.7–95.1	3.1	19	17.98	21.18	27.78
90%	24	in-region	SCP-T4	84.2	65.7–95.3	7.7	50	13.39	14.38	33.82
90%	24	in-region	ACI-CQR	90	89.8–90.1	0.1	0	18.37	98.41	104.47
90%	24	in-region	ACI-CQR-V	89.9	89.4–90.4	0.2	0	17.75	21.03	27.81
90%	24	unseen-region	QR	87.3	76.1–96.1	3.8	29	17.15	20.14	28.26

Nominal	h (h)	Setting	Method	Mean coverage %	Min–max coverage %	Mean	cov. dev.	pp	BAs under-covered >5 pp	Median hourly width %
90%	24	unseen-region	SCP	88.9	71.4–99.3	5.6	29	20.14	20.88	31.88
90%	24	unseen-region	CQR	89.2	78.6–97.3	3.5	21	18.14	21.18	28.15
90%	24	unseen-region	SCP-T4	85.3	60.8–95.9	6.3	40	14.97	15.68	34.15
90%	24	unseen-region	ACI-CQR	90	89.8–90.2	0.1	0	18.6	82.58	88.55
90%	24	unseen-region	ACI-CQR-V	89.9	89.4–90.5	0.3	0	17.81	21.32	28.24
50%	1	in-region	QR	41.9	28.2–57.8	8.9	69	1.31	1.48	3.78
50%	1	in-region	SCP	48	21.3–75.8	10.7	40	1.49	1.54	3.79
50%	1	in-region	CQR	48.5	32.2–67.6	7.2	35	1.55	1.73	3.75
50%	1	in-region	SCP-T4	46.9	32.2–62.0	5.6	35	1.44	1.53	3.83
50%	1	in-region	ACI-CQR	49.9	49.6–50.1	0.1	0	1.64	1.84	3.75
50%	1	in-region	ACI-CQR-V	49.8	48.9–50.3	0.2	0	1.63	1.83	3.75
50%	1	unseen-region	QR	41.4	27.2–58.7	9.8	65	1.35	1.52	3.9
50%	1	unseen-region	SCP	46.3	20.3–73.6	10.5	50	1.49	1.54	3.91
50%	1	unseen-region	CQR	47.7	31.0–66.9	8.1	42	1.58	1.77	3.88
50%	1	unseen-region	SCP-T4	47.3	34.0–64.5	5.6	35	1.52	1.6	3.9
50%	1	unseen-region	ACI-CQR	49.9	49.5–50.2	0.1	0	1.69	1.93	3.87
50%	1	unseen-region	ACI-CQR-V	49.8	48.9–50.4	0.2	0	1.69	1.92	3.87
50%	24	in-region	QR	47.6	36.5–55.5	3.9	25	6.31	7.5	15.27
50%	24	in-region	SCP	48.3	29.8–74.1	7.4	38	6.11	6.31	15.6
50%	24	in-region	CQR	48.8	37.5–56.8	3.8	19	6.5	7.69	15.28
50%	24	in-region	SCP-T4	47.4	20.9–68.8	9	48	5.67	5.84	15.78
50%	24	in-region	ACI-CQR	49.9	49.6–50.2	0.1	0	6.62	7.66	15.34
50%	24	in-region	ACI-CQR-V	49.7	48.6–50.6	0.4	0	6.59	7.54	15.28
50%	24	unseen-region	QR	47.1	36.3–58.2	4.5	29	6.23	7.4	15.64
50%	24	unseen-region	SCP	47.9	29.9–73.1	7.6	35	6.08	6.28	15.89
50%	24	unseen-region	CQR	48.8	37.6–60.7	4.2	25	6.49	7.68	15.64
50%	24	unseen-region	SCP-T4	48.9	23.3–65.0	7	33	6	6.15	16.07
50%	24	unseen-region	ACI-CQR	49.9	49.6–50.2	0.1	0	6.75	7.69	15.68
50%	24	unseen-region	ACI-CQR-V	49.7	48.5–50.7	0.4	0	6.63	7.54	15.64

2 S2. Sensitivity of ACI to the step size (post hoc, test set)

This table was not used for any selection.

Table S2. POST-HOC test-set sensitivity of ACI-CQR (90%, unseen BAs) to the step size γ (not used for any selection).

h (h)	γ	Mean coverage %	Min coverage %	Median hourly width % of mean load	Median interval score %	Capped hours %
1	0.001	90	89.5	4.88	7.13	0
1	0.002	90	89.7	4.9	7.13	0
1	0.005	90	89.9	4.9	7.4	0.09
1	0.01	90	89.9	4.93	7.71	0.46
24	0.001	89.9	89.4	17.81	28.24	0.13
24	0.002	89.9	89.6	18.05	29.35	0.97
24	0.005	90	89.8	18.6	88.55	4.37
24	0.01	90	89.9	19.51	272.91	12.25

3 S3. Missing-input robustness, all scenarios

Table S3. Robustness to missing input observations, all scenarios (test period; common scored hours).

h (h)	Scenario	Strategy	Model	Input missing %	nMAE % median	nMAE % mean	Min. scored share of hours
1	BLOCK_168h	locf	LGB_G_IN	9.7	2.42	2.51	0.969
1	BLOCK_168h	locf	LGB_G_LRO	9.7	2.46	2.56	0.969
1	BLOCK_168h	native	LGB_G_IN	9.7	1.95	2.08	0.969
1	BLOCK_168h	native	LGB_G_IN_AUG	9.7	1.94	2.01	0.969
1	BLOCK_168h	native	LGB_G_LRO	9.7	1.99	2.11	0.969
1	BLOCK_168h	seasonal	LGB_G_IN	9.7	1.87	1.93	0.969
1	BLOCK_168h	seasonal	LGB_G_LRO	9.7	1.9	1.98	0.969
1	BLOCK_24h	locf	LGB_G_IN	9.5	2.52	2.56	1
1	BLOCK_24h	locf	LGB_G_LRO	9.5	2.56	2.6	1
1	BLOCK_24h	native	LGB_G_IN	9.5	1.89	1.98	1
1	BLOCK_24h	native	LGB_G_IN_AUG	9.5	1.71	1.81	1
1	BLOCK_24h	native	LGB_G_LRO	9.5	1.97	2.02	1
1	BLOCK_24h	seasonal	LGB_G_IN	9.5	1.91	2.02	1
1	BLOCK_24h	seasonal	LGB_G_LRO	9.5	1.93	2.07	1
1	MCAR_10	locf	LGB_G_IN	10.2	1.68	1.66	1
1	MCAR_10	locf	LGB_G_LRO	10.2	1.7	1.71	1
1	MCAR_10	native	LGB_G_IN	10.2	1.84	1.84	1
1	MCAR_10	native	LGB_G_IN_AUG	10.2	1.32	1.41	1
1	MCAR_10	native	LGB_G_LRO	10.2	1.93	1.93	1
1	MCAR_10	seasonal	LGB_G_IN	10.2	2.19	2.26	1
1	MCAR_10	seasonal	LGB_G_LRO	10.2	2.22	2.31	1
1	MCAR_20	locf	LGB_G_IN	20.2	2.21	2.2	1
1	MCAR_20	locf	LGB_G_LRO	20.2	2.24	2.24	1
1	MCAR_20	native	LGB_G_IN	20.2	2.56	2.51	1
1	MCAR_20	native	LGB_G_IN_AUG	20.2	1.52	1.62	1
1	MCAR_20	native	LGB_G_LRO	20.2	2.7	2.64	1
1	MCAR_20	seasonal	LGB_G_IN	20.2	3.19	3.27	1
1	MCAR_20	seasonal	LGB_G_LRO	20.2	3.25	3.31	1
1	MCAR_40	locf	LGB_G_IN	40.1	3.68	3.68	1
1	MCAR_40	locf	LGB_G_LRO	40.1	3.68	3.7	1
1	MCAR_40	native	LGB_G_IN	40.1	4.04	4.11	1

h (h)	Scenario	Strategy	Model	Input missing %	nMAE % median	nMAE % mean	Min. scored share of hours
1	MCAR_40	native	LGB_G_IN_AUG	40.1	2.29	2.34	1
1	MCAR_40	native	LGB_G_LRO	40.1	4.22	4.31	1
1	MCAR_40	seasonal	LGB_G_IN	40.1	5.14	5.28	1
1	MCAR_40	seasonal	LGB_G_LRO	40.1	5.19	5.31	1
1	MCAR_5	locf	LGB_G_IN	5.2	1.37	1.44	1
1	MCAR_5	locf	LGB_G_LRO	5.2	1.41	1.48	1
1	MCAR_5	native	LGB_G_IN	5.2	1.47	1.53	1
1	MCAR_5	native	LGB_G_IN_AUG	5.2	1.25	1.33	1
1	MCAR_5	native	LGB_G_LRO	5.2	1.56	1.6	1
1	MCAR_5	seasonal	LGB_G_IN	5.2	1.66	1.76	1
1	MCAR_5	seasonal	LGB_G_LRO	5.2	1.7	1.8	1
1	clean	native	LGB_G_IN	0.2	1.14	1.23	1
1	clean	native	LGB_G_IN_AUG	0.2	1.17	1.26	1
1	clean	native	LGB_G_LRO	0.2	1.17	1.28	1
24	BLOCK_168h	locf	LGB_G_IN	9.7	5.79	5.85	0.969
24	BLOCK_168h	locf	LGB_G_LRO	9.7	5.82	5.88	0.969
24	BLOCK_168h	native	LGB_G_IN	9.7	5.07	5.34	0.969
24	BLOCK_168h	native	LGB_G_IN_AUG	9.7	5.03	5.3	0.969
24	BLOCK_168h	native	LGB_G_LRO	9.7	5.08	5.38	0.969
24	BLOCK_168h	seasonal	LGB_G_IN	9.7	5.07	5.34	0.969
24	BLOCK_168h	seasonal	LGB_G_LRO	9.7	5.09	5.38	0.969
24	BLOCK_24h	locf	LGB_G_IN	9.5	5.81	5.82	1
24	BLOCK_24h	locf	LGB_G_LRO	9.5	5.83	5.85	1
24	BLOCK_24h	native	LGB_G_IN	9.5	5.02	5.3	1
24	BLOCK_24h	native	LGB_G_IN_AUG	9.5	4.94	5.22	1
24	BLOCK_24h	native	LGB_G_LRO	9.5	5.12	5.33	1
24	BLOCK_24h	seasonal	LGB_G_IN	9.5	5.13	5.39	1
24	BLOCK_24h	seasonal	LGB_G_LRO	9.5	5.21	5.42	1
24	MCAR_10	locf	LGB_G_IN	10.2	4.81	5.09	1
24	MCAR_10	locf	LGB_G_LRO	10.2	4.91	5.12	1
24	MCAR_10	native	LGB_G_IN	10.2	5.05	5.33	1
24	MCAR_10	native	LGB_G_IN_AUG	10.2	4.72	5.03	1
24	MCAR_10	native	LGB_G_LRO	10.2	5.14	5.36	1
24	MCAR_10	seasonal	LGB_G_IN	10.2	5.06	5.41	1
24	MCAR_10	seasonal	LGB_G_LRO	10.2	5.16	5.45	1
24	MCAR_20	locf	LGB_G_IN	20.2	5.05	5.28	1
24	MCAR_20	locf	LGB_G_LRO	20.2	5.13	5.31	1
24	MCAR_20	native	LGB_G_IN	20.2	5.62	5.77	1
24	MCAR_20	native	LGB_G_IN_AUG	20.2	4.87	5.12	1
24	MCAR_20	native	LGB_G_LRO	20.2	5.69	5.79	1
24	MCAR_20	seasonal	LGB_G_IN	20.2	5.57	5.89	1
24	MCAR_20	seasonal	LGB_G_LRO	20.2	5.65	5.91	1
24	MCAR_40	locf	LGB_G_IN	40.1	5.98	5.95	1
24	MCAR_40	locf	LGB_G_LRO	40.1	5.99	5.98	1
24	MCAR_40	native	LGB_G_IN	40.1	6.93	6.88	1
24	MCAR_40	native	LGB_G_IN_AUG	40.1	5.4	5.54	1
24	MCAR_40	native	LGB_G_LRO	40.1	7.02	6.9	1
24	MCAR_40	seasonal	LGB_G_IN	40.1	6.66	6.91	1
24	MCAR_40	seasonal	LGB_G_LRO	40.1	6.69	6.93	1
24	MCAR_5	locf	LGB_G_IN	5.2	4.72	5.02	1
24	MCAR_5	locf	LGB_G_LRO	5.2	4.82	5.06	1
24	MCAR_5	native	LGB_G_IN	5.2	4.84	5.14	1

h (h)	Scenario	Strategy	Model	Input missing %	nMAE % median	nMAE % mean	Min. scored share of hours
24	MCAR_5	native	LGB_G_IN_AUG	5.2	4.68	4.99	1
24	MCAR_5	native	LGB_G_LRO	5.2	4.94	5.18	1
24	MCAR_5	seasonal	LGB_G_IN	5.2	4.86	5.19	1
24	MCAR_5	seasonal	LGB_G_LRO	5.2	4.96	5.22	1
24	clean	native	LGB_G_IN	0.2	4.64	4.96	1
24	clean	native	LGB_G_IN_AUG	0.2	4.65	4.97	1
24	clean	native	LGB_G_LRO	0.2	4.75	5	1

4 S4. All paired comparisons of point accuracy

Table S4. All paired comparisons of per-BA nMAE (test period).

h (h)	A	B	Median rel. change in nMAE (A vs B) %	95% CI (regions bootstrap)	A better (share of BAs)	Wilcoxon p (Holm, all)	BAs with A sig. better (block bootstrap)	BAs with B sig. better
1	LightGBM, global (unseen region, zero-shot)	LightGBM, global (in-region)	3.4	[2.5, 4.6]	0	5.1e-13	0.0	0.98
1	LightGBM, global (in-region)	LightGBM, local	-9.2	[-11.7, -7.3]	0.92	2.1e-08	0.83	0.04
1	LightGBM, global (unseen region, zero-shot)	LightGBM, local	-5.4	[-8.1, -3.7]	0.77	0.00052	0.75	0.12
1	LightGBM, global (unseen region, zero-shot)	Seasonal naive (24 h)	-79.6	[-81.7, -77.5]	1	5.1e-13	1.0	0.0
1	LightGBM, global (in-region)	Ridge, global (in-region)	-19.3	[-22.3, -15.5]	1	5.1e-13	1.0	0.0
1	Ridge, global (unseen region)	Ridge, global (in-region)	0.5	[0.2, 0.6]	0.23	0.0083	–	–
1	LightGBM, global (unseen region, zero-shot)	Ridge, global (unseen region)	-14.8	[-19.7, -11.5]	0.98	2.1e-11	–	–
1	LightGBM, local	Seasonal naive (24 h)	-78.1	[-79.2, -76.4]	1	5.1e-13	–	–
6	LightGBM, global (unseen region, zero-shot)	LightGBM, global (in-region)	1.9	[1.6, 2.3]	0	5.1e-13	0.0	0.9
6	LightGBM, global (in-region)	LightGBM, local	-5.7	[-7.1, -4.0]	0.88	3.9e-08	0.79	0.04
6	LightGBM, global (unseen region, zero-shot)	LightGBM, local	-3.1	[-5.6, -1.8]	0.71	0.0067	0.54	0.15
6	LightGBM, global (unseen region, zero-shot)	Seasonal naive (24 h)	-35.3	[-37.2, -32.6]	0.98	6.4e-13	0.98	0.02
6	LightGBM, global (in-region)	Ridge, global (in-region)	-15.8	[-18.1, -14.1]	1	5.1e-13	1.0	0.0
6	Ridge, global (unseen region)	Ridge, global (in-region)	0.2	[0.1, 0.2]	0.25	0.00065	–	–
6	LightGBM, global (unseen region, zero-shot)	Ridge, global (unseen region)	-13	[-15.5, -12.3]	0.98	3.6e-12	–	–

h (h)	A	B	Median rel. change in nMAE (A vs B) %	95% CI (regions bootstrap)	A better (share of BAs)	Wilcoxon p (Holm, all)	BAs with A sig. better (block bootstrap)	BAs with B sig. better
6	LightGBM, local	Seasonal naive (24 h)	-32.5	[-35.3, -29.9]	1	5.1e-13	–	–
24	LightGBM, global (unseen region, zero-shot)	LightGBM, global (in-region)	0.5	[0.4, 0.7]	0.06	2.5e-11	0.0	0.52
24	LightGBM, global (in-region)	LightGBM, local	-1.4	[-2.2, -0.8]	0.71	0.0029	0.35	0.06
24	LightGBM, global (unseen region, zero-shot)	LightGBM, local	-0.8	[-1.6, 0.2]	0.62	0.21	0.31	0.17
24	LightGBM, global (unseen region, zero-shot)	Seasonal naive (24 h)	-12	[-15.2, -10.5]	1	5.1e-13	0.94	0.0
24	LightGBM, global (in-region)	Ridge, global (in-region)	-5.5	[-7.1, -5.2]	1	5.1e-13	0.98	0.0
24	Ridge, global (unseen region)	Ridge, global (in-region)	0.1	[0.0, 0.2]	0.25	0.00021	–	–
24	LightGBM, global (unseen region, zero-shot)	Ridge, global (unseen region)	-5.4	[-6.5, -4.6]	1	5.1e-13	–	–
24	LightGBM, local	Seasonal naive (24 h)	-11.7	[-14.3, -10.1]	0.98	6.4e-13	–	–

5 S5. Adaptation at additional onboarding origins (post-review)

Table S5. Median nMAE (%) for unseen BAs at three additional onboarding origins (evaluation: the following 365 days).

h	origin	LGB_FT_k13	LGB_FT_k4	LGB_G_LRO	LGB_L_k13	LGB_L_k4
6	2024-04-01	3.83	3.85	3.7	7.8	7.66
6	2024-07-01	3.68	3.82	3.7	5.23	6.99
6	2024-10-01	3.63	3.73	3.63	5.52	6.09
24	2024-04-01	4.88	4.95	4.67	7.36	8.26
24	2024-07-01	5.35	5.97	4.75	6.21	7.85
24	2024-10-01	5.1	5.34	4.81	5.9	6.68

6 S6. Spatially blocked leave-region-out folds (post-review)

The nine blocks are the EIA reporting regions NW, CAL, SW, FLA and CAR, plus four merged groups of adjacent single-BA eastern regions: TEX+CENT, MIDW+MIDA, SE+TEN and NE+NY. When a block is held out, all its BAs are excluded from fitting, early stopping and calibration.

Table S6. Point accuracy with spatially blocked leave-region-out folds (test period).

h (h)	Method	nMAE % median [IQR]	MASE median
1	LightGBM, global (in-region)	1.14 [0.87, 1.51]	0.194
1	LightGBM, global (unseen region, zero-shot)	1.17 [0.90, 1.55]	0.201
1	LightGBM, global (spatially blocked, zero-shot)	1.27 [1.01, 1.61]	0.219
1	LightGBM, local	1.28 [1.01, 1.61]	0.22
1	Seasonal naive (24 h)	5.48 [4.57, 6.81]	1.009

h (h)	Method	nMAE % median [IQR]	MASE median
24	LightGBM, global (in-region)	4.64 [3.88, 5.87]	0.884
24	LightGBM, global (unseen region, zero-shot)	4.75 [3.91, 5.91]	0.888
24	LightGBM, global (spatially blocked, zero-shot)	4.81 [4.02, 6.03]	0.901
24	LightGBM, local	4.72 [3.94, 5.94]	0.893
24	Seasonal naive (24 h)	5.48 [4.57, 6.81]	1.009

7 S7. Excluded BAs (post-review)

Table S7. Zero-shot forecasts for the five BAs excluded by the selection criteria.

h (h)	BA	Mean MW	Zero-shot nMAE %	Local nMAE %	SN24 nMAE %
1	HST	73	1.99	1.67	7.1
1	PSEI	2836	0.71	0.91	4.77
1	SEC	233	3.74	4.16	11.13
1	SPA	22	24.38	21.04	30.12
1	WAUW	91	3.3	2.91	6.45
24	HST	73	7.02	6.69	7.1
24	PSEI	2836	4.01	4.1	4.77
24	SEC	233	10.29	10.54	11.13
24	SPA	22	25.3	23.4	30.12
24	WAUW	91	6.4	6.44	6.45

8 S8. Conditional coverage (post-review)

Table S8. Conditional coverage of 90% intervals by season and by within-BA load decile (0 = lowest).

Setting	h (h)	Method	Conditioning	Level	Mean cov. over BAs %	Min cov. over BAs %
unseen (random folds)	24	CQR	season	DJF	86.8	71.5
unseen (random folds)	24	CQR	season	JJA	89.9	73.5
unseen (random folds)	24	CQR	season	MAM	90	80.2
unseen (random folds)	24	CQR	season	SON	90	80.3
unseen (random folds)	24	CQR	decile	0	88.2	70
unseen (random folds)	24	CQR	decile	1	90	76.9
unseen (random folds)	24	CQR	decile	2	91.1	81.8
unseen (random folds)	24	CQR	decile	3	91.3	83.1
unseen (random folds)	24	CQR	decile	4	91.1	78.7
unseen (random folds)	24	CQR	decile	5	89.7	76.9
unseen (random folds)	24	CQR	decile	6	88.5	76.6
unseen (random folds)	24	CQR	decile	7	87.6	76.6
unseen (random folds)	24	CQR	decile	8	87.3	74.1
unseen (random folds)	24	CQR	decile	9	87	65.9

Setting	h (h)	Method	Conditioning	Level	Mean cov. over BAs %	Min cov. over BAs %
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	season	DJF	88.3	81.7
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	season	JJA	90	84.9
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	season	MAM	91.1	84.8
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	season	SON	90	85.2
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	0	89.3	81.4
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	1	90.9	87.8
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	2	91.7	87.9
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	3	91.9	88.1
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	4	91.6	87.8
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	5	90.3	86.4
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	6	89	85.8
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	7	88.1	85
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	8	87.9	80.4
unseen (random folds)	24	ACI-CQR-S (gamma source-selected)	decile	9	87.9	72
unseen (random folds)	1	CQR	season	DJF	87.8	70.5
unseen (random folds)	1	CQR	season	JJA	88.5	66.2
unseen (random folds)	1	CQR	season	MAM	88.1	68.5
unseen (random folds)	1	CQR	season	SON	88	67.5
unseen (random folds)	1	CQR	decile	0	87.9	69.4
unseen (random folds)	1	CQR	decile	1	87.9	70.6
unseen (random folds)	1	CQR	decile	2	88.1	67.2
unseen (random folds)	1	CQR	decile	3	87.8	69.5
unseen (random folds)	1	CQR	decile	4	88	68.6
unseen (random folds)	1	CQR	decile	5	88.2	70.3
unseen (random folds)	1	CQR	decile	6	88.4	68.9
unseen (random folds)	1	CQR	decile	7	88.3	67.3
unseen (random folds)	1	CQR	decile	8	88.3	65.7
unseen (random folds)	1	CQR	decile	9	88.2	64.9
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	season	DJF	89.9	89.1

Setting	h (h)	Method	Conditioning	Level	Mean cov. over BAs %	Min cov. over BAs %
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	season	JJA	90	89
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	season	MAM	90.2	89.6
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	season	SON	89.8	88.9
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	0	90.2	85.2
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	1	90	87.6
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	2	90.3	87.1
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	3	90	87.5
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	4	90.1	87.5
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	5	90	87.7
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	6	90.1	88.1
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	7	89.7	86.9
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	8	89.6	85.5
unseen (random folds)	1	ACI-CQR-S (gamma source-selected)	decile	9	89.7	83.3
unseen (spatial blocks)	24	CQR	season	DJF	87.1	70.9
unseen (spatial blocks)	24	CQR	season	JJA	90.3	73.6
unseen (spatial blocks)	24	CQR	season	MAM	90.4	82
unseen (spatial blocks)	24	CQR	season	SON	90.3	81
unseen (spatial blocks)	24	CQR	decile	0	88.6	73.1
unseen (spatial blocks)	24	CQR	decile	1	90.5	78.5
unseen (spatial blocks)	24	CQR	decile	2	91.5	79.7
unseen (spatial blocks)	24	CQR	decile	3	91.6	82.7
unseen (spatial blocks)	24	CQR	decile	4	91.4	79.2
unseen (spatial blocks)	24	CQR	decile	5	90	77.3
unseen (spatial blocks)	24	CQR	decile	6	88.9	76.7
unseen (spatial blocks)	24	CQR	decile	7	87.9	76.6
unseen (spatial blocks)	24	CQR	decile	8	87.7	73.7
unseen (spatial blocks)	24	CQR	decile	9	87.2	64.6
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	season	DJF	88.5	81.6
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	season	JJA	90.1	85.4

Setting	h (h)	Method	Conditioning	Level	Mean cov. over BAs %	Min cov. over BAs %
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	season	MAM	91.1	84.4
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	season	SON	89.8	84.8
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	0	89	82.1
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	1	91	88
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	2	91.7	89.2
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	3	91.9	87.1
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	4	91.7	87.5
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	5	90.3	86.9
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	6	89.2	85.8
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	7	88.2	84.8
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	8	87.9	80.8
unseen (spatial blocks)	24	ACI-CQR-S (gamma source-selected)	decile	9	87.9	71.4

9 S9. Ridge penalty grid (post-review)

Table S9. Ridge penalty grid on the validation (odd ISO) weeks, h = 24 h.

Ridge penalty λ	Validation median nMAE %
0.01	5.059
0.1	5.059
1	5.059
10	5.059
100	5.059
1000	5.059

10 S10. Adaptation compute cost

Table S10. Adaptation fit time per target BA (s).

model	h	fit_seconds_per_region
LGB_L_k1	1	0.062
LGB_FT_k1	1	0.839
LGB_L_k4	1	0.157
LGB_FT_k4	1	1.017
LGB_L_k13	1	0.247
LGB_FT_k13	1	1.383
LGB_L_k1	6	0.06
LGB_FT_k1	6	0.297
LGB_L_k4	6	0.168
LGB_FT_k4	6	0.403
LGB_L_k13	6	0.256
LGB_FT_k13	6	0.531
LGB_L_k1	24	0.062

model	h	fit_seconds_per_region
LGB_FT_k1	24	0.151
LGB_L_k4	24	0.182
LGB_FT_k4	24	0.232
LGB_L_k13	24	0.255
LGB_FT_k13	24	0.295